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Faults in Determining Nitrogenase Activity

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With 4 figures and 1 table

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Abstract

The effects of three treatments performed before the acetylene reduction assay, i.e. shoot removal, shoot removal plus root shaking, and no disturbance (intact plants), and of the length of the incubation period (10, 30 or 60 min) on nitrogenase activity were investigated in pea (*Pisum sativum* L.). The shoot removal plus root shaking treatment decreased nitrogenase activity to a greater extent than shoot removal. Nitrogenase activity was significantly higher when measured after a 10-min incubation period than after a 30- or 60-min incubation period. In another experiment, plants disturbed 15 min before the assay had a higher activity than plants disturbed 75 or 135 min before the assay. Similar results were obtained when activity was expressed as total nitrogenase activity and as specific nitrogenase activity.

Key words: nitrogenase activity — pea — root shaking — shoot removal — treatment of plants before assay

Introduction

The standard acetylene reduction assay (Hardy et al. 1968, 1973, Jonsson 1988) for measuring the nitrogenase activity of nodulated legumes involves the incubation of nodulated roots or detached nodules in a closed vessel. In the standard acetylene reduction assay, the shoot is removed and the root system is disturbed to obtain nodules. The removal of the shoot (Hardy et al. 1973, Minchin et al. 1986, Herdina and Silsbury 1990), root shaking (Minchin et al. 1986) and soaking of roots in water (Mague and Burris 1972, Trinick et al. 1976) reduce nitrogenase activity. The effect of plant disturbance on nitrogenase activity is expected to vary depending upon the length of time that elapses between sampling and measurement of nitrogenase activity. This may be an important consideration in field experiments where sampling may take place over a long period.

It is possible to measure the nitrogenase activity of intact plants using a flow-through gas system (Cooper et al. 1982, Minchin et al. 1983). This technique avoids any acetylene-induced decline in nitrogenase activity (Minchin et al. 1983), but can only be used for pot experiments, as for field experiments one has to disturb the root system to recover roots and nodules from the soil. The acetylene reduction assay, however, is very popular with researchers and has been used to measure nitrogenase activity in groundnut (Daimon and Yoshioka 2001), faba bean (Filek et al. 2000), soybean (Sridhara et al. 1995) and lentil (Singh et al. 1993).

The experiments reported in this paper were conducted to study the effects of various plant disturbance treatments before the assay on nitrogenase activity in peas, to reflect the types of disturbance that normally occur not only in field experiments but also in pot experiments when the plants are grown in soil. Experiment 1 was conducted to answer the following questions: (i) Does plant disturbance decrease nitrogenase activity? (ii) What is the effect of shoot removal, or shoot removal combined with root shaking? (iii) What is the effect of the length of the incubation period on nitrogenase activity under disturbed and undisturbed plant conditions? Experiment 2 was conducted to study the effect of shoot removal and the length of time between sampling and measurement on nitrogenase activity, with the aim of determining ways to improve the use of the standard acetylene reduction assay for field-grown plants.

Materials and Methods

With the objective of studying the effects of various plant disturbance treatments before the assay on nitrogenase activity in pea, two pot experiments were conducted at the University of Wales, Bangor, Gwynedd, UK. The plants were grown in a growth room at day/night temperatures of 18/9 °C, with a photoperiod of 16 h and relative humidity of 70 %. The growth room had photosynthetic photon flux densities of between 150 and 220 μmol m^{-2} s^{-1} at the tops of plants during the growing season of the plants. However, plants in experiment 2 received lower light intensities than these for about 3 weeks.

In experiment 1, two seeds of cv. Rex were sown in 800-cm^3 plastic pots (height 12 cm and diameter 9.2 cm) containing perlite. In experiment 2, eight seeds of cv. Rex in four hills (two seeds per hill) were sown in perlite in 5-l capacity plastic pots having a surface area of 342 cm^2. In experiment 1, before sowing seeds were soaked in liquid *Rhizobium leguminosarum* culture for 1 h. In this experiment, sowing was shallow (depth 1.5 cm) and during the process of germination seeds were pushed upwards. These were covered with extra perlite. In experiment 2, sowing was therefore deeper (depth 4 cm). Up to the germination stage, the pots were watered daily with tap water. After germination, Long Ashton N-free nutrient solution (Hewitt 1966) was used to feed the plants 3 days a week (Mondays, Wednesdays and Fridays) and on other days of the week the plants were watered with tap water. In experiment 1, only one plant per pot was retained, whereas in experiment 2 four plants per pot (one plant per hill) were retained. The plants were inoculated with 4 cm^3 of 1-week-old liquid culture of *Rhizobium leguminosarum* strain RCR 1045, prepared in yeast mannitol broth (Vincent 1970, Somasegaran and Hoben 1994). The plants were supported by sticks to keep them upright.

In both the experiments, all the plants were uniform and did not receive any treatment until the time of the acetylene reduction assay. In experiment 1, at the time of measurement of nitrogenase activity the following three plant disturbance treatments were applied to the 65–67-day-old plants, when they were at the flowering and early pod formation stage (stage code 204; Knott 1987), 1 h before injecting acetylene into the pots: (i) shoot removal – the shoot portion was cut at the perlite/air interface and the roots were not disturbed; (ii) shoot removal + root shaking – the shoot was cut and the roots were shaken to remove the perlite attached to them, then washed and dried with tissue paper and put back into the empty washed and dried pots, and (iii) no disturbance (intact) – neither shoot nor root system was disturbed. Nitrogenase activity was measured on intact plants by sealing the pots in which the plants were growing. The stem of the plants was passed through the hole in the partially cut lid, which was replaced at the time of the acetylene reduction assay. To determine the effect of the length of the incubation period on the nitrogenase activity of differentially disturbed plants, after injecting acetylene gas, samples of 0.5 cm^3 were removed after incubation periods of 10, 30 and 60 min.

In experiment 2, the following six (3 × 2) treatments were applied to the 65-day-old pea plants which were at the flowering stage (between stage codes 203 and 204; Knott 1987) at the time of measurement of nitrogenase activity. (i) The plants were removed from perlite 135, 75 and 15 min before the assay and the roots were shaken to remove the attached perlite. (ii) After subjecting the plants to treatment (i) above, the shoots were left intact in half of the plants and were removed in the other plants 15 min before the assay. Two plants for each treatment were transferred into 800-cm^3 plastic pots. The stems of the plants that retained the shoot portion were passed through the holes in the partially cut lids. In this case the plants (shoots) were supported in an upright position by a thin stick attached to the outside of the pot. The drainage holes as well as the holes in the lids were plugged with plasticine to make the pots gas-tight. The plants experienced the same temperature and light (photoperiod as well as light intensity) during the growing period, treatment period and assay period.

In the shoot removal + root shaking treatment in experiment 1 and all the treatments in experiment 2, the pots did not contain any perlite and thus the whole pot volume of 800 cm^3 was available for diffusion of gasses. However, for the shoot removal and no disturbance treatments in experiment 1, the 800-cm^3 pots also contained perlite and the empty pore space was estimated by a displacement method, to determine the volume available for dilution of gases, which was about 400 cm^3. Acetylene (10 % v/v) was injected into the pots with a syringe through a small hole in the lid, and as the pressure built up the same amount of air was allowed to escape through another needle which was placed in the lid for this purpose. After injecting acetylene, both the needles were removed and the pots were made fully gas-tight immediately by plugging these small holes with plasticine. Gas samples of 0.5 cm^3 were taken with 1-cm^3 syringes after incubation periods of 10, 30 and 60 min in experiment 1 and after 30 min in experiment 2. These samples were run in a gas chromatograph for measurement of ethylene concentration. Nitrogenase activity was calculated using the method of Jonsson (1988). Nitrogenase activity was expressed as μmol C_2H_4 $plant^{-1}$ h^{-1} (total nitrogenase activity) or μmol C_2H_4 g^{-1} nodule fresh weight h^{-1} or μmol C_2H_4 g^{-1} nodule dry weight h^{-1} (both known as specific nitrogenase activity). After measurement of nitrogenase activity, the plants were harvested. Although visually very uniform plants were selected for these studies, some parameters, such as nodulation, and root and shoot growth, were recorded to confirm the uniformity of plants used in various treatments.

Experiments 1 and 2 had six and four replicates, respectively, and both experiments were conducted in a randomized complete block design. All data were analysed by analysis of variance (ANOVA) using Minitab statistical package version 10.51 (Anonymous 1992). The significance of the F ratio was tested at the 5 % probability level and the means were compared by calculating a least significance difference by the t-distribution method.

Results

The plants used in the various plant disturbance treatments in experiments 1 and 2 had similar numbers and fresh and dry weights of nodules (Table 1), tap root length, plant height, and root and shoot dry weights (data not shown).

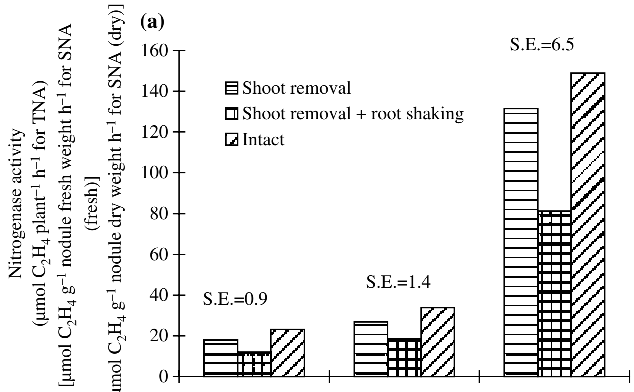
In experiment 1, plant disturbance treatments affected nitrogenase activity significantly (Fig. 1a). Intact plants had the highest nitrogenase activity, which was significantly higher than that of the shoot removal and shoot removal + root shaking treatments, the latter two treatments also differing significantly from each other. The data showed that plant disturbance decreased nitrogenase activity, the effect of shoot removal + root shaking being far greater than that of shoot removal only. Averaged over all incubation periods, shoot removal and shoot removal + root shaking decreased total nitrogenase activity by 23 and 48 %, respectively, compared to the intact plants. A similar trend was recorded for the effect of the various plant disturbance treatments on nitrogenase activity when data were expressed on a per plant, per gram nodule fresh weight or per gram nodule dry weight basis.

Incubation period influenced nitrogenase activity significantly when data were expressed on a per plant basis or a per unit nodule fresh or dry weight basis (Fig. 1b). Total nitrogenase activity was significantly higher when measured after a 10-min incubation period than after a 30- or 60-min incubation period. The data showed a similar trend when nitrogenase activity was expressed as g^{-1} nodule fresh weight or g^{-1} nodule dry weight.

Table 1: Nodulation of plants used for the various plant disturbance treatments

Treatment	No. of nodules plant^{-1}	Nodule weight plant^{-1} (mg)	
		Fresh	Dry
Experiment 1			
Shoot removal	107.0	678	140
Shoot removal + root shaking	105.8	658	151
Undisturbed (intact)	105.1	681	157
L.S.D. 5 %	ns	ns	ns
Experiment 2			
Shoot treatment			
Retained	93.4	408	99
Removed	94.0	414	101
L.S.D. 5 %	ns	ns	ns
Time between sampling and measurement (min)			
15	92.6	420	100
75	94.4	414	103
135	94.0	399	97
L.S.D. 5 %	ns	ns	ns

The interaction between plant disturbance treatments and incubation period was significant at the 5 % probability level when nitrogenase activity was expressed as per plant or g^{-1} nodule fresh weight (Fig. 2a,b). This interaction was significant only at the 10 % probability level when nitrogenase activity was calculated as g^{-1} nodule dry weight (Fig. 2c). Differences in nitrogenase activity between treatments decreased as the length of incubation period increased. The no disturbance (intact) and shoot removal treatments were significantly different for the 10-min incubation period but not for the 60-min incubation period. Similarly, the shoot removal + root shaking treatment produced a significantly lower nitrogenase activity than the treatment of shoot removal only for the 10-min incubation period but not for the 60-min



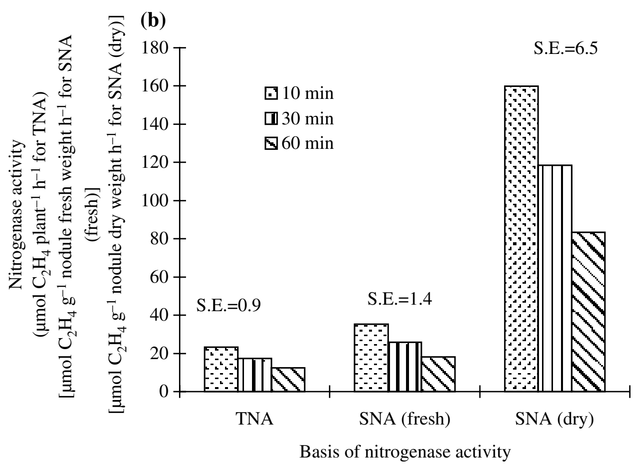


Fig. 1: Effect of (a) plant disturbance treatments and (b) length of incubation period on total nitrogenase activity (TNA) and specific nitrogenase activity (SNA) in pea (values of S.E.M. are at 40 d.f.)

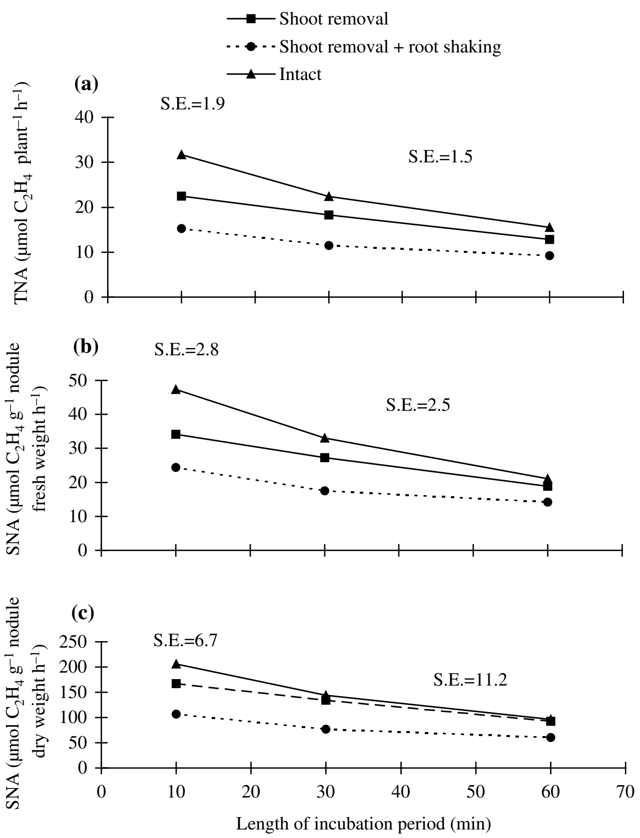


Fig. 2: Interaction effects of plant disturbance treatments and length of incubation period on (a) total nitrogenase activity (TNA) and (b, c) specific nitrogenase activity (SNA) in pea (values of S.E.M. are at 40 d.f.)

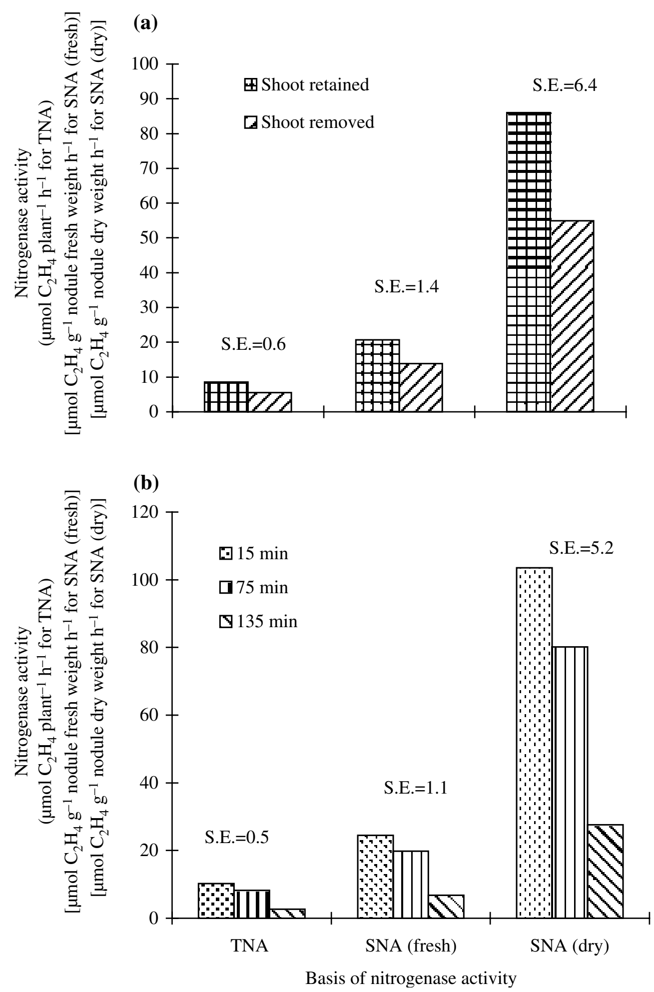


Fig. 3: Effect of (a) shoot removal and (b) time between sampling and measurement of nitrogenase activity on total nitrogenase activity (TNA) and specific nitrogenase activity (SNA) in pea (values of S.E.M. are at 15 d.f.)

incubation period. However, the duration of the incubation period did not alter the rank order of the different treatments. As the shorter incubation period (10 min) provided greater differences in nitrogenase activity, these data were analysed separately and the results (Fig. 2) showed significant differences between intact plants and plants of both the disturbance treatments (shoot removal and shoot removal + root shaking) as well as between plants of the two disturbance treatments.

In experiment 2, the plants with intact shoots had significantly higher nitrogenase activity than the plants in which the shoot was removed 15 min before the assay (Fig. 3a). Removal of shoots decreased nitrogenase activity irrespective of whether it was expressed as per plant, g^{-1} nodule fresh weight or g^{-1} nodule dry weight, the decrease being 35, 35 and 36 %, respectively, relative to intact plants.

The plants disturbed only 15 min before the assay showed the highest nitrogenase activity, which was significantly higher than that recorded for plants disturbed 75 or 135 min before the assay (Fig. 3b). This statement is true for activity expressed in any of the three ways. When nitrogenase activity was expressed per plant, the decreases in activity with increase in root disturbance time from 15 to 75 min and from 75 to 135 min before the assay were 20 and 68 %, respectively. Similar trends were observed when activity was expressed g^{-1} nodule fresh weight or g^{-1} nodule dry weight.

Although all interactions were non-significant at the 5 % probability level, the adverse effect of shoot removal on nitrogenase activity was much greater for root disturbance 75 min before the assay than for disturbance of 15 or 135 min before the assay (Fig. 4). Increasing the time between harvesting of plants and measurement of nitrogenase activity had a much larger effect on plants with detached shoots than on intact plants, and this

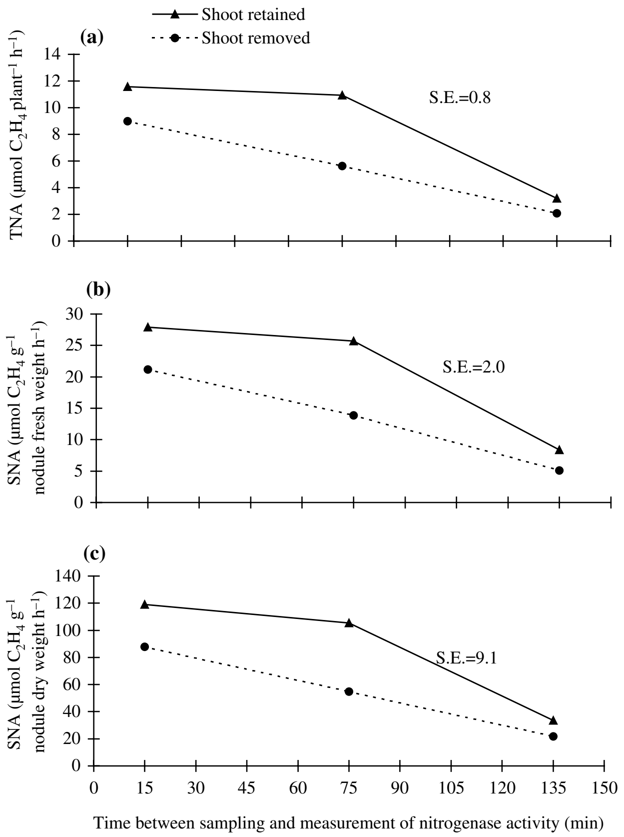


Fig. 4: Interaction effects of shoot removal and time between sampling and measurement of activity on (a) total nitrogenase activity (TNA) and (b, c) specific nitrogenase activity (SNA) in pea (values of S.E.M. are at 15 d.f.)

trend was the same for activity expressed in any of the three ways.

Discussion

The rates of nitrogenase activity were higher in all the treatments after an incubation period of 10 min, and as the length of incubation period increased a decrease in nitrogenase activity was observed in all the treatments (Figs 1 and 2). In intact plants, the decrease in nitrogenase activity with longer incubation periods is assumed to be attributable only to acetylene, as acetylene has been found to decrease nitrogenase activity (Minchin et al. 1983, Witty and Minchin 1988). In the shoot removal and shoot removal + root shaking treatments, the decrease was smaller than that for undisturbed plants. This smaller decrease may be attributable to a smaller acetylene-induced decline, as the acetylene-induced decline decreases as the effect of plant disturbance increases (Minchin et al. 1986). In both experiments, shoot removal resulted in decreased nitrogenase activity over undisturbed (Fig. 1) or shoot-retaining plants (Fig. 3). As the degree of plant disturbance increased, nitrogenase activity decreased. This may explain why the shoot removal + root shaking treatment decreased nitrogenase activity more than did shoot removal only. Minchin et al. (1986) also found similar results with these types of plant disturbance treatments in soybean.

Disturbance can decrease nitrogenase activity as a result of physical disturbance of the nodules (Minchin et al. 1986), limitation of nodule function through effects on carbohydrate supply (Minchin et al. 1986, Vessey et al. 1988a), or a feedback mechanism produced by the accumulation of fixed nitrogen (as reported in stem-girdled plants by Streeter 1993). The length of time between sampling and measurement is an important factor which can affect nitrogenase activity by influencing carbohydrate limitation of nodules. When shoots are detached, nitrogenase activity eventually declines as a result of a decrease in carbohydrate supply to the nodules (Vessey et al. 1988a). However, in another study, Vessey et al. (1988b) found that even when the shoot was removed nitrogenase activity continued for up to 20–30 min before inhibition took place. Walsh et al. (1987) found that the length of time for which nitrogenase activity can continue before inhibition, after cutting the shoot supply of carbohydrates, depends on the carbohydrate status of the nodule and root tissue. This may explain why the plants with shoots removed had higher rates of nitrogenase activity for root disturbance 15 min before the assay than for root disturbance 75 or 135 min before the assay. In the intact shoot treatment, as the supply of carbohydrates from the shoot to the nodules was maintained, higher values of nitrogenase activity were obtained that for the shoot removal treatment, even for longer periods of 75 and 135 min between sampling and measurement of activity (Fig. 4).

Minchin et al. (1986) found that the resistance to oxygen diffusion of nodules on detopped/shaken roots is substantially larger than that of nodules on intact roots, and suggested that the decrease in nitrogenase activity of detopped/shaken roots is a reflection of decreased oxygen supply to the bacteroid. This increase in diffusion resistance may be attributable to reduced photosynthate supply following detopping, as Minchin et al. (1985) found similar responses with carbohydrate-stressed nodules in plants subjected to 24 h of shoot darkening.

The undisturbed, intact plants had higher nitrogenase activity than plants in the shoot removal or shoot removal + root shaking treatment. This suggests that the method of measuring nitrogenase activity on intact plants growing in a porous medium, which is expected to contain some water, is better than methods (e.g. the shoot removal + root shaking treatment, in which effects of water were minimized) in which the plants are subjected to various degrees of disturbance. The rank order of the treatments was the same across incubation periods, but for longer incubation periods and longer times between sampling and measurement the values of nitrogenase activity were lower, and this may reduce the ability to detect differences between treatments.

Conclusions

The treatment of shoot removal + root shaking decreased nitrogenase activity more than shoot removal only, and intact plants had the highest rates of nitrogenase activity. So, in pot experiments, nitrogenase activity should be measured on intact plants, and to avoid or minimize an acetylene-induced decline nitrogenase activity should be measured after the shortest possible incubation period (i.e. 10 min). In field experiments, however, excavation of roots is unavoidable. In these experiments the length of time between sampling and measurement of nitrogenase activity should be kept to a minimum. Furthermore, retaining shoots at the time of measurement of nitrogenase activity of nodulated roots can result in higher activity values. However, further research is needed to determine the conditions under which these plants should be kept during the assay period, as in some parts of the world under natural conditions very high temperatures may result in wilting of the plants.

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